

Ym1 protein crystals promote type 2 immunity

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Abstract

Background

Spontaneous protein crystallization is a rare event, yet needle-shaped protein crystals are frequently found in eosinophil-rich inflammation. In humans, Charcot-Leyden crystals (CLCs) are made from galectin-10 (Gal10) protein, an abundant protein in eosinophils.

Aim

Whereas mice do not encode Gal10 in their genome, they do form pseudo-CLCs, made from the chitinase-like proteins Ym1 and/or Ym2, encoded by *Chil3* and *Chil4* and made by myeloid and epithelial cells respectively. Here, we investigated the biological effects of pseudo-CLCs since their function is currently unknown.

Methods and results

We produced recombinant Ym1 crystals which were shown to have identical crystal packing and structure by X-ray crystallography as *in vivo* native crystals derived from murine lung. When administered to the airways of mice, crystalline but not soluble Ym1 stimulated innate and adaptive immunity and acted as a type 2 immune adjuvant for eosinophilic inflammation via triggering of dendritic cells (DCs).

Conclusion

Murine Ym1 protein crystals found at sites of eosinophilic inflammation reinforce type 2 immunity and could serve as a surrogate model for studying the biology of human CLCs.

Protein crystallization is exceedingly rare, yet here, Heyndrickx et al. show that Ym1 crystals made from the chitinase like protein Chil3 promote type 2 immunity in mice.

eLife assessment

This study presents **compelling** observations of the ability of Ym1 crystals to promote type 2 immunity *in vivo*, in mice, that advances our knowledge of how the abundantly expressed chitinase-like protein family may function, with new methodology, new crystal structures that are more rigorous than what is currently known and very **convincing** immunological data. The significance of the study is **important**, providing an exciting opportunity to mechanistically study the impact of protein crystals *in vivo* in relation to human disease. This work will be of broad interest to immunologists and researchers working on type 2 inflammatory disease, in the lung.

Introduction

Protein crystals have been described already 150 years ago in eosinophilic inflammatory lesions in humans by Charcot and von Leyden^{1,2,3}. Most proteins can only function properly in the soluble state and spontaneous protein crystallization *in vivo* remains a rare and largely unsolved scientific enigma^{4,5}.

Protein crystals have also been documented in mouse tissues, almost exclusively in the context of type 2 inflammation, rich in eosinophils, Th2 cells and alternatively activated macrophages⁶⁻²². Unstained crystals are colorless, but in histochemical stains they stain brightly with eosin, consistent with their protein content¹⁴. Before their biochemical nature was elucidated, they were therefore often referred to as eosinophilic crystals⁶. These murine crystals are frequently observed in aged mice and genetically altered mice on a C57BL/6 or Sv129 background, and in the context of eosinophil-rich asthma and parasitic infections^{6,23}. Although eosinophilic crystals have been documented in the skin, biliary tree, stomach, lymph nodes, spleen and bone marrow of mice, it is clear that the principal site of crystal accumulation is in the lung. In several descriptions, accumulation of eosinophilic crystals was associated with severe lung inflammation and the disease was called eosinophilic crystalline pneumonia, acidophilic macrophage pneumonia or crystalline pneumonitis^{7,17,23,24}. The accumulated crystals are variable in size and shape ranging from needle-like crystals to rectangular plates¹⁴. Their resemblance to the needle-shaped human Charcot-Leyden crystals (CLCs) frequently resulted in the incorrect identification of these murine crystals as CLCs^{24,25}. However, only in 2000, almost 100 years after they were first described, Guo *et al.* purified crystals from the bronchoalveolar lavage (BAL) fluid of viable motheaten (*me^v/me^v*) mice and mass spectrometry analysis identified the chitinase-like protein (CLP) Ym1 as the sole protein component⁶, thereby providing an unambiguous annotation of their content and distinguishing them from CLCs. Human CLCs are comprised of the eosinophil-derived galectin-10 (Gal10) protein, which is not encoded in the mouse genome^{26,27}. Ym1 is produced by a subset of circulating pro-repair monocytes, by alternatively activated “M2” macrophages, alveolar macrophages, and by neutrophils²⁸⁻³⁰. In hyaline gastric lesions of aged 129S4/SvJae and B6.129 CYP1A2 null mice, pseudo-CLCs were found to consist of the closely related CLP Ym2, which is a protein mainly made by epithelial cells under the influence of IL-13^{14,25,31,32}. Ym1 (encoded by the *Chil3* gene) and Ym2 (encoded by the *Chil4* gene) are CLPs belonging to the glycoside hydrolase family 18

(GH18) and are unique to mice, with no known human orthologues ³³[\[33\]](#). Their potential receptors and natural binding partners are largely unknown, as are the exact functions of these still very enigmatic proteins.

We previously reported that recombinant Gal10 crystals produced *in vitro* showed crystal packing and structure identical to CLCs derived from patient mucus. When administered to the airways of mice, Gal10 crystals stimulated innate and adaptive immunity and acted as a type 2 adjuvant ³⁴[\[34\]](#). Since mice do not encode the *LGALS10* gene, Gal10 has the potential to act as a neo-antigen in this setting. Here, we tested the *in vivo* relevance of pseudo-CLCs since these are made endogenously in mice at sites of eosinophilic inflammation, albeit by macrophages and epithelial cells. Furthermore, given the high sequence identity of Ym1 and Ym2 we hypothesized whether Ym2 might also crystallize *in vivo* analogous to Ym1. Recombinant *in vitro* made Ym1 and Ym2 crystals were biosimilar to crystals isolated from mice, and when administered to the airways of mice, crystalline Ym1, but not soluble Ym1 protein stimulated innate and adaptive immunity and acted as an adjuvant for Th2 driven eosinophilic airway inflammation and immunoglobulin secretion.

Methods

Mice

Female wild type (WT) C57BL/6 mice were purchased from Janvier Labs (Saint-Berthevin, France), housed under specific pathogen-free conditions and used between 6 and 8 weeks of age. Investigators were not blinded to group allocation during experiments. No animals were excluded from the analysis. All experiments were approved by the animal ethics committee at Ghent University and were in accordance with Belgian animal protection law.

Production of recombinant Ym1 and Ym2 protein and crystals

A synthetic codon-optimized DNA sequence (Genscript) encoding murine Ym1 (Uniprot ID O35744, residues 22 – 398) was cloned in the pCAGG mammalian expression vector in frame with the mouse IgH signal peptide (MGWSCIIFFLVATATGVHS). A synthetic codon-optimized DNA sequence (Genscript) encoding murine Ym2 (Uniprot Q91Z98, residues 22 - 402) was cloned in the pTwist-CMV-BetaGlobin mammalian expression vector (Twist Biosciences), in frame with the mouse IgH signal peptide (MGWSCIIFFLVATATGVHS), and an N-terminal hexahistidine-tag followed by a Tobacco Etch Virus (TEV)-protease cleavage site.

Ym1 and Ym2 were produced in FreeStyle 293-F cells in suspension. Cells were transfected with 1 µg/ml DNA using 2-fold excess of linear polyethylenimine (PEI; Polysciences). Conditioned medium containing the secreted recombinant protein was collected 4 days post transfection, filtered and dialyzed to 20 mM Tris pH 8.0. Ym1 was purified via Q Sepharose anion exchange chromatography. Ym1 was further purified to a pure solution via size-exclusion chromatography, while Ym2 was purified from the clarified conditioned medium via immobilized metal affinity chromatography (IMAC) and size-exclusion chromatography with phosphate-buffered saline (PBS) buffer. Endotoxin levels were determined with an Endosafe-PTS system (Charles River) to be lower than 1 EU/mg of recombinant protein. To form recombinant Ym1 crystals, recombinant Ym1 protein (3 to 4 mg/ml) was incubated with 1M Na-acetate buffer pH 4.6 with a buffer:target protein ratio of 1:10 (v/v). During a 24h incubation the protein solution was agitated a few times by inverting it five times, after which the solution turned cloudy due to the formation of protein crystals. The crystal-containing suspension was spun down (400g for 5 min) and washed three times with sterile and endotoxin-free PBS before the crystals were resuspended in 1 ml of sterile and endotoxin-free PBS. The total protein concentration of this solution was determined by solubilizing 20 µl of the crystal suspension in an equal volume of 6M guanidinium hydrochloride and measuring the absorbance at 280 nm with a Thermo Scientific NanoDrop 2000. The calculated extinction coefficient for Ym1 was 77155 M⁻¹ cm⁻¹.

Crystallographic structure determination of *ex vivo* and recombinant Ym1 and Ym2 crystals

Single *ex vivo* Ym1 and Ym2 crystals were harvested from the BAL fluid of me^V/me^V mice and C57BL/6J-Tg(CAG-Gal10)Bla mice by using mounted cryo-loops. Before being cryo-cooled in liquid nitrogen, crystals were cryo-protected by a brief soak in PBS supplemented with 35% (v/v) ethylene glycol (Ym1) or 35 % glycerol (Ym2). Recombinant Ym1 was crystallized in condition A7 of the Hampton Research Crystal Screen HT (0.1 M sodium cacodylate pH 6.5, 1.4 M sodium acetate) via vapor diffusion sitting drop crystallization experiments, and crystals were cryo-protected with mother liquor supplemented with 30% (v/v) glycerol. Following overnight incubation with TEV-protease, recombinant Ym2 was crystallized via vapor diffusion sitting drop crystallization experiments using 0.1 M sodium acetate pH 6.0 and 200 mM CaCl₂ as mother liquor, and crystals were cryoprotected with 30% (v/v) PEG-400. Diffraction experiments at 100 K were conducted on beamlines Proxima 1 (SOLEIL synchrotron, Saint-Aubin, France), P14 of Petra III (DESY, Hamburg, Germany) and X06SA (PXI) of the Swiss Light Source (Paul Scherrer Institute, Villigen, Switzerland). All data were integrated and scaled using the XDS suite³⁵ and the CCP4 suite³⁶. Molecular replacement was executed using Phaser³⁷ employing the structure of Ym1 (pdb: 1vfb) as a search model. Further rebuilding of the model and refinement was performed in COOT³⁸ and autoBuster³⁹. Map and model validation was monitored throughout the refinement procedure using COOT, the CCP4 package and PDB_REDO server⁴⁰.

In vivo models

To analyze the innate immune responses, WT C57BL/6 mice were injected intratracheally (i.t.) with 100 µg Ym1 crystals or 100 µg soluble Ym1 diluted in 80 µl of PBS. After 6 and 24 hours, mice were euthanized by CO₂ inhalation or an overdose of pentobarbital (KELA Laboratoria) and BAL fluid and lungs were collected.

To analyze dendritic cell (DC) activation, WT C57BL/6 mice were injected i.t. with 5 µg Alexa Fluor 488-labeled ovalbumin (OVA-AF488) mixed with 100 µg Ym1 crystals or with 100 µg soluble Ym1 diluted in 80 µl of PBS. After 24 hours, mice were euthanized by CO₂ inhalation and mediastinal lymph nodes (mLNs) were collected.

To analyze the adaptive cellular immune responses, CD4⁺ OT-II KN2 cells were purified from CD45.1.2 OT-II KN2 donor mice with the MojoSort™ Mouse CD4 T Cell Isolation Kit (Biolegend) and labeled with cell proliferation dye eFluor 450 (0.01 mM; Thermo Fisher Scientific) for 10 min in the dark at 37°C. Next, 1 × 10⁶ of these labeled cells (in 100 µl of PBS) were injected intravenously (i.v.) into WT C57BL/6 mice and subsequently, mice were injected i.t. with 10 µg ovalbumin (OVA, Worthington) or with 10 µg OVA mixed with 100 µg of Ym1 crystals or 100 µg soluble Ym1 (in 80 µl of PBS). For this experiment, OVA was pre-filtered through a 0.22 µm filter. On day 4 and 7, mice were euthanized by CO₂ inhalation and mLNs and lungs were collected.

To analyze the adaptive humoral immune responses, WT C57BL/6 mice were injected intraperitoneally (i.p.) on day 0 with 0.5 mg Ym1 crystals or 0.5 mg soluble Ym1, on day 7 with 0.5 mg Ym1 crystals and on day 14 with 0.5 mg Ym1 crystals or 10 µg Sol Ym1 all diluted in 500 µl PBS. On day 21 mice were euthanized with an overdose of pentobarbital (KELA Laboratoria) and blood was collected.

To analyze the type 2 adjuvanticity of Ym1 crystals, WT C57BL/6 mice were sensitized i.t. on day 0 and day 1 with 100 µg OVA (Worthington), 100 µg OVA mixed with 100 µg of Ym1 crystals or 100 µg OVA mixed with 100 µg soluble Ym1 diluted in 80 µl PBS. On days 11 to 13, all mice were challenged daily intranasally (i.n.) with 20 µg of OVA (Worthington) diluted in 40 µl of PBS. One day after the last challenge, on day 14, mice were euthanized with an overdose of pentobarbital (KELA Laboratoria) and blood, BAL fluid and mLNs were collected.

All i.t. and i.n. treatments were given after mice were anesthetized with isoflurane (2 liters/min, 2 to 3%; Abbott Laboratories).

Blood was obtained from the iliac vein and centrifuged at 10000 rpm for 10 min at room temperature (RT). Serum was collected and stored at -20°C until further use (serum antibody ELISA). BAL was performed by injecting PBS containing 0.01 mM EDTA (Lonza) through a tracheal canula. Subsequently, BAL fluid was centrifuged (400g for 5 min at 4°C). The supernatant was collected and stored at -20°C until further use (cytokine ELISA) and the pellet was used for ImageStream or flow cytometry. To obtain lung and mLN single-cell suspensions for flow cytometry, lungs and mLNs were first cut with a scissor and then digested at 37°C for 30 or 15 min respectively in RPMI-1640 (ThermoFisher Scientific) containing Liberase TM (20 $\mu\text{g}/\text{ml}$; Merck) and deoxyribonuclease (DNase) I (0.01 U/ μl ; Merck). The resultant cell suspensions were filtered through a 70 μm filter. To obtain mLN single-cell suspensions for flow cytometry or mLN restimulation cultures, mLNs were smashed on a 70 μm filter. To measure pro-inflammatory cytokines, lungs were snap frozen in liquid nitrogen and homogenized with a TissueLyser II from Qiagen in tissue lysis buffer [40 mM tris-HCL (pH 6.8), 275mM NaCl and 20% glycerol (Sigma-Aldrich)]. One tablet of PhosSTOP (Sigma-Aldrich) and one tablet of Complete ULTRA tablets, Mini, Easypack (Sigma-Aldrich) were added per 10 ml of tissue lysis buffer. After homogenization, 2% IGEPAL CA-630 (U.S. Biological) was added. Subsequently, the samples were rotated for 30 min and centrifuged (20000g for 10 min at 4°C). Supernatants were stored at -20°C . The total protein concentration was measured with the NanoOrange protein quantitation kit from Thermo Fisher Scientific.

Flow cytometry

Prior to staining for flow cytometry, single-cell suspensions were depleted of red blood cells (RBCs) by using RBC lysis buffer [0.15 M NH_4Cl (Merck), 1 mM KHCO_3 (Merck), and 0.1 mM $\text{Na}_2\text{-EDTA}$ (Merck) in MilliQ H_2O] produced in-house. For all flow experiments, single cell suspensions were incubated with the eBioscience™ Fixable Viability Dye eFluor™ 506 (ThermoFisher Scientific) to identify dead cells. Fc Block 2.4.G2 (Bioceros) was used to block aspecific antibody binding. Cell surface markers were stained for 30 min at 4°C in the dark. 123count eBeads™ Counting Beads (ThermoFisher Scientific) were added to each sample to determine absolute cell numbers. Settings were calibrated using UltraComp eBead™ Compensation Beads (ThermoFisher Scientific). Data were acquired on an LSR Fortessa with FACS Diva software (BD Biosciences) and were analyzed with FlowJo software (Tree Star).

For the analysis of innate immune responses, lung single-cell suspensions were stained for flow cytometry using FITC-conjugated anti-Ly-6C (AL-21) (BD Biosciences), PE-conjugated anti-Siglec-F (E50-2440) (BD Biosciences), PE-Cy5-conjugated anti-CD3e (145-2c11) (Thermo Fisher Scientific), PE-Cy5-conjugated anti-CD19 (1D3) (Thermo Fisher Scientific), PE-Cy7-conjugated anti-CD11c (N418) (Thermo Fisher Scientific), BD Horizon V450-conjugated anti-CD11b (M1/70) (BD Biosciences), Brilliant Violet 605-conjugated anti-CD45 (30-F11) (BD Biosciences), Alexa Fluor 647-conjugated anti-CD64 (X54-5/7.1) (BD Biosciences), Alexa Fluor 700–conjugated anti-Ly6G (1A8) (BD Biosciences) and APC-eFluor 780–conjugated anti-I-A/I-E (M5/114.15.2) (Thermo Fisher Scientific).

For the analysis of DC activation, mLN single-cell suspensions were stained for flow cytometry using PerCP-Cy5.5-conjugated anti-CD80 (16-10A1) (BioLegend), eFluor 450-conjugated anti-CD11c (N418) (ThermoFisher Scientific), Brilliant Violet 650-conjugated anti-XCR1 (ZET) (BioLegend), Brilliant Violet 711-conjugated anti-CD64 (X54-5/7.1) (BioLegend), Alexa Fluor 647-conjugated anti-CD172a (P84) (BioLegend), Alexa Fluor 700-conjugated anti-Ly-6C (AL-21) (BD Biosciences), APC-eFluor 780-conjugated anti-MHCII (M5/114.15.2) (ThermoFisher Scientific), PE-conjugated anti-CD88 (20/70) (BioLegend), biotin-conjugated anti-CCR7 (4B12) (ThermoFisher Scientific) + PE-CF594-conjugated streptavidin (BD Biosciences), PE-Cy5-conjugated anti-CD3e (145-2c11)

(ThermoFisher Scientific), PE-Cy5-conjugated anti-CD19 (1D3) (Thermo Fisher Scientific), PE-Cy7-conjugated anti-CD86 (PO3) (BioLegend), BUV395-conjugated anti-CD11b (M1/70) (BD Biosciences) and BUV737-conjugated anti-CD26 (H194-112) (BD Biosciences).

For T cell response analysis, mLN and lung single-cell suspensions were stained using FITC-conjugated anti-Va2 TCR (B20.1) (BD Biosciences), Brilliant Violet 605-conjugated anti-CD45.1 (A20) (BioLegend), RedFluor710-conjugated anti-CD44 (IM7) (Thermo Fisher Scientific), APC-eFluor780-conjugated anti-CD25 (PC61.5) (Thermo Fisher Scientific), PE-conjugated anti-CD62L (MEL-14) (BD Biosciences), biotin-conjugated anti-hCD2 (RPA-2.10) (BioLegend) + PE-CF594-conjugated streptavidin (BD Biosciences), PE-Cy7-conjugated CD3e (145-2C11) (BioLegend), BUV395-conjugated anti-CD4 (GK1.5) (BD Biosciences) and BUV737-conjugated anti-CD45.2 (104) (BD Biosciences).

BAL cells were stained with PE-conjugated anti-Siglec F (E50-2440) (BD Biosciences), PE-Cy5-conjugated anti-CD3e (145-2c11) (ThermoFisher Scientific), PE-Cy5-conjugated anti-CD19 (1D3) (Thermo Fisher Scientific), PE-Cy7-conjugated anti-CD11c (N418) (ThermoFisher Scientific), BD Horizon V450-conjugated anti-CD11b (M1/70) (BD Biosciences), Alexa Fluor 700-conjugated anti-Ly6G (1A8) (BD Biosciences) and APC-eFluor 780-conjugated anti-MHCII (M5/114.15.2) (Thermo Fisher Scientific).

ImageStream

For ImageStream analysis, the pellets obtained after centrifugation of the BAL fluid were pooled and stained for 30 min at 4°C with unconjugated polyclonal goat anti-mouse Ym1/2 (AF2446, R&D Systems; 1:100 diluted in PBS). After washing, the pellet was stained for 30 min at 4°C in the dark with Alexa Fluor 647-conjugated polyclonal donkey anti-goat IgG (ThermoFisher Scientific; 1:1000 diluted in PBS). Fc Block 2.4.G2 (Bioceros) was used to block aspecific antibody binding. Data were acquired on an Amnis ImageStreamX Mk II with INSPIRE™ software (Luminex) and were analyzed with IDEAS® software (Luminex).

Enzyme-linked immunosorbent assay (ELISA)

Interleukin (IL)-1 β , IL-5, IL-6, IL-10, IL-13, IL-33, tumor necrosis factor alpha (TNF α) and C-C motif chemokine ligand 2 (CCL2) concentrations were evaluated using the Ready-SET-Go!™ ELISA kits (ThermoFisher Scientific). CCL24 concentrations were evaluated using the Mouse CCL24/Eotaxin-2/MPIF-2 DuoSet ELISA kit (R&D Systems). Serum Ym1-specific immunoglobulin M (IgM) and OVA-specific immunoglobulin G1 (IgG1) antibody levels were determined in flat-bottom 96-well plates (Greiner Bio-One), coated overnight with 0.1 mg/ml soluble Ym1 in PBS or with 0.1 mg/ml OVA (Worthington) in 0.1 M sodium carbonate buffer (pH 9.5), respectively. After washing and blocking for 1 hour, the samples diluted in PBS were added in duplicate and incubated for 2.5 hours at RT. After washing, biotinylated anti-mouse IgG1 or IgM detecting antibody (0.5 mg/ml; BD Biosciences) plus streptavidin-HRP reagent (1:250; BD Biosciences) was added and incubated for 1 hour at RT. After another wash step, 1 $\times$ TMB substrate solution (Thermo Fisher Scientific) was added and to stop the reaction, 2.5N H₂SO₄ was used. Finally, the absorbance was read at 450 nm (and 650 nm as reference) with a Perkin Elmer multilabel counter and data were collected with Wallac 1420 Manager software (PerkinElmer).

Statistical analysis

Experiments shown in **Figure 2** [↗](#) were analyzed with an ordinary two-way ANOVA test with Tukey's multiple comparisons test. For all experiments in **Figure 3** [↗](#), **4** [↗](#) and **5** [↗](#), the normality of each group was first checked by using the Shapiro-Wilk statistical test. Parametric data were analyzed with an ordinary one-way ANOVA test with Šidák's multiple comparison correction and nonparametric data were analyzed with a Kruskal-Wallis test with Dunn's multiple comparison correction. All statistical tests were performed in GraphPad Prism software (GraphPad Software).

Supplementary Online Material

This paper has an online data supplement. Table S1 describes the key crystallographic data of *ex vivo* isolated Ym1 and Ym2 crystals for comparison with *in vitro* generated recombinant Ym1 and Ym2 crystals.

Results

Recombinant Ym1 and Ym2 crystals are structurally similar to native crystals *ex vivo*

Individual crystals were harvested from the BAL fluid of me^V/me^V mice (carrying a defective protein tyrosine phosphatase SHP-1 gene) developing crystalline pneumonia [6](#) and C57BL/6J-Tg (*Cagg-Gal10*)^{Bla} mice (transgenically overexpressing Gal10 from the ubiquitous *Cagg* promoter, manuscript in preparation), sensitized and challenged with the allergen house dust mite (HDM). Crystallographic studies using microfocus synchrotron X-rays showed that the crystals obtained from the BAL fluid of me^V/me^V mice were Ym1 crystals, in accordance with a previous report [6](#). We had hypothesized that the crystals obtained from the BAL fluid of C57BL/6J-Tg(CAG-Gal10)^{Bla} mice were CLCs containing Gal10. However, crystallographic characterization of these crystals revealed that they are Ym2 crystals. In parallel, we produced recombinant Ym1 and Ym2 protein in FreeStyle 293-F cells. Spontaneous crystallization was aided by low pH and low temperature (4°C) (**Fig. 1 A, B**). Native *in vivo* isolated Ym1 crystals and recombinant Ym1 crystals (**Fig. 1 C**) as well as native *in vivo* isolated Ym2 crystals and recombinant Ym2 crystals (**Fig. 1 D**) were crystallographically equivalent, sharing the same hexagonal space group (P21) and unit-cell parameters. Moreover, X-ray crystallographic studies showed that a native *in vivo* isolated Ym1 crystal and a native *in vivo* isolated Ym2 crystal are structurally similar to each other, which is not so surprisingly as Ym1 (*Chil3*) and Ym2 (*Chil4*) are two closely related proteins sharing 95% nucleotide sequence identity and 91% amino acid sequence identity (**Fig. 1 E, F, G**). Crystallographic data and refinement statistics can be found in Table S1. Thus, recombinant Ym1 and Ym2 protein crystals can be generated *in vitro* and are biosimilar to native Ym1 and Ym2 crystals.

Ym1 crystals activate innate immunity

We next addressed whether Ym1 crystals could have pro-inflammatory effects in the lungs. Naïve WT C57BL/6 mice received an i.t. injection of 100 µg of endotoxin-low Ym1 crystals or control soluble Ym1 (**Fig. 2 A**). After 6 and 24 hours, there was a prominent influx of Ly6C⁺ monocytes and eosinophils into the airways of mice receiving Ym1 crystals but not in those receiving soluble Ym1 or PBS. Neutrophil numbers were also increased in the Ym1 crystal group, but this failed to reach statistical significance (**Fig. 2 B**). This influx was accompanied by the production of the innate pro-inflammatory cytokines IL-6 and TNFα in BAL fluid (**Fig. 2 C**) and the inflammasome-dependent cytokine IL-1β and the alarmin IL-33 in lung tissue only in the group receiving crystalline Ym1 (**Fig. 2D**). Other cytokines like IL-1α and GM-CSF were not consistently induced in any of the groups (data not shown). The monocyte chemoattractant CCL2 and the eosinophil chemoattractant CCL24 (Eotaxin-2) were found enhanced in lung tissue of the crystal-treated group (**Fig. 2 E**), particularly at early time points, suggesting these paracrine signals were driving the cellular influx induced by crystals. Cellular influx and production of these cytokines and chemokines was also observed in earlier work where recombinant Gal10 crystals or uric acid (UA) crystals were injected into the lungs of mice, suggesting this is a generic response to crystals of different chemical composition^{34,41}.

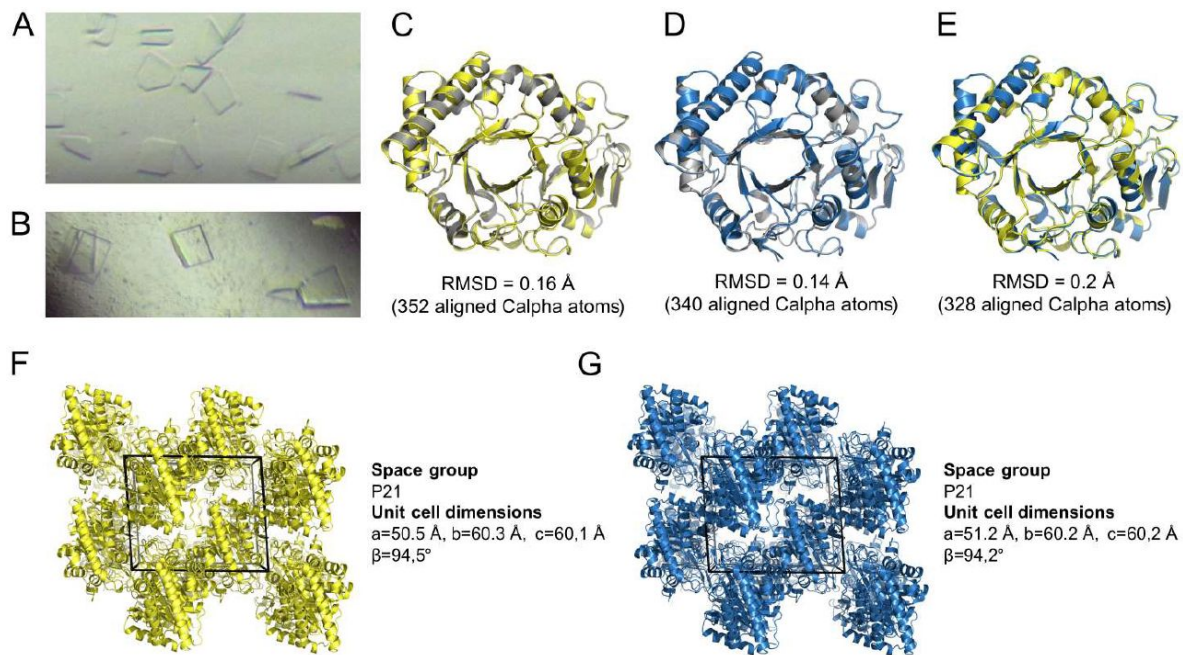


Figure 1

Recombinant Ym1 and Ym2 crystals are structurally similar to *in vivo* Ym1 and Ym2 crystals and to each other.

A, B) Light microscopy pictures of recombinant Ym1 (A) and Ym2 (B) crystals. **C-E)** Structural superposition of the X-ray structures determined from a native *in vivo* grown Ym1 crystal isolated from the BAL fluid of me^V/me^V mice (C+E) (yellow), a native *in vivo* grown Ym2 crystal isolated from the BAL fluid of a C57BL/6J-Tg(CAG-Gal10)Bla mouse (D+E) (blue) and a recombinant Ym1 (C) and Ym2 (D) crystal (grey). The low root-mean-square deviations (RMSDs) establish structural similarity. **F, G)** Crystal packing of *ex vivo* Ym1 (F) and Ym2 (G) crystals revealing their crystallographic equivalence.

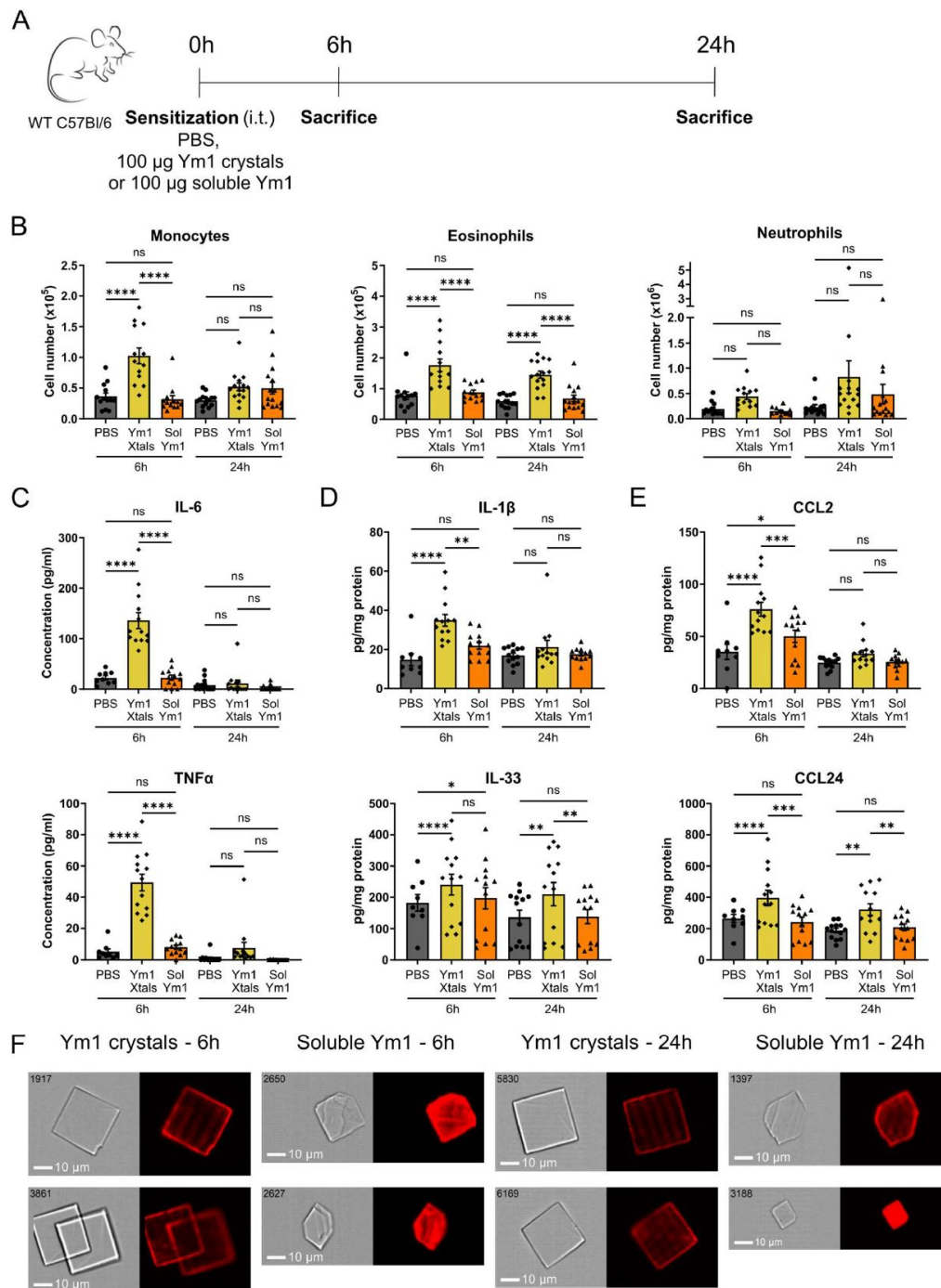


Figure 2

Ym1 crystals activate innate immunity.

A) Scheme representing the innate immune response experiment. **B)** Flow cytometric analysis of the immune cell influx in the lungs of PBS, Ym1 crystal and soluble Ym1 treated mice measured 6 and 24 hours after i.t. injection. Data are pooled from 2 (for 6h time point) or 3 (for 24h time point) independent experiments with $n = 13, 13, 13, 14, 15$ and 15 , respectively. **C)** Concentration of IL-6 and TNFα in the BAL fluid measured by ELISA. **D, E)** Concentration of IL-1β, IL-33 (D), CCL2 and CCL24 (E) per milligram of protein of dispersed lung tissue measured by ELISA. Data are pooled from 2 independent experiments with $n = 13$ per group. Data are shown as mean $\pm$ SEM. ns $P \geq 0.05$; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$. **E)** ImageStream pictures of the BAL fluid of Ym1 crystal and soluble Ym1 treated mice 6 and 24 hours after i.t. injection. Samples were stained with an antibody recognizing Ym1/2 (red).

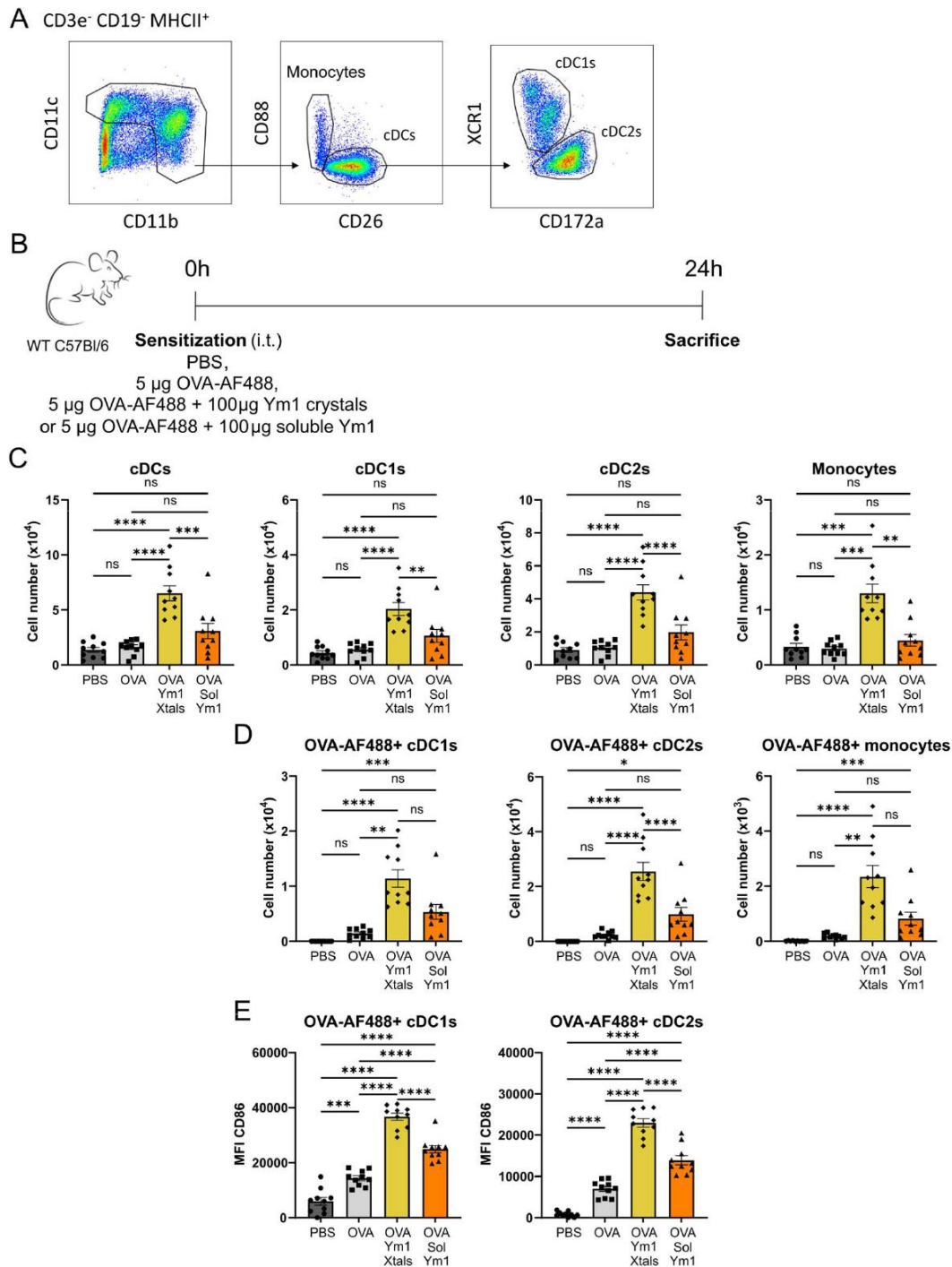


Figure 3

Ym1 crystals activate DCs in the lymph nodes.

A) Gating strategy to distinguish different subsets of DCs and recruited monocytes. **B)** Scheme representing the DC activation experiment. **C, D)** Flow cytometric analysis of the DC and monocyte influx in the mLNs of PBS, OVA-AF488 alone, OVA-AF488 plus Ym1 crystals and OVA-AF488 plus soluble Ym1 treated mice measured 24 hours after i.t. injection. **E)** Median Fluorescence Intensity (MFI) of activation marker CD86 on OVA-AF488+ cDC1s and cDC2s of PBS, OVA-AF488 alone, OVA-AF488 plus Ym1 crystals and OVA-AF488 plus soluble Ym1 treated mice measured 24 hours after i.t. injection. Data are pooled from 2 independent experiments with $n = 10$ mice per group. Data are shown as mean $\pm$ SEM. ns $P \geq 0.05$; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.

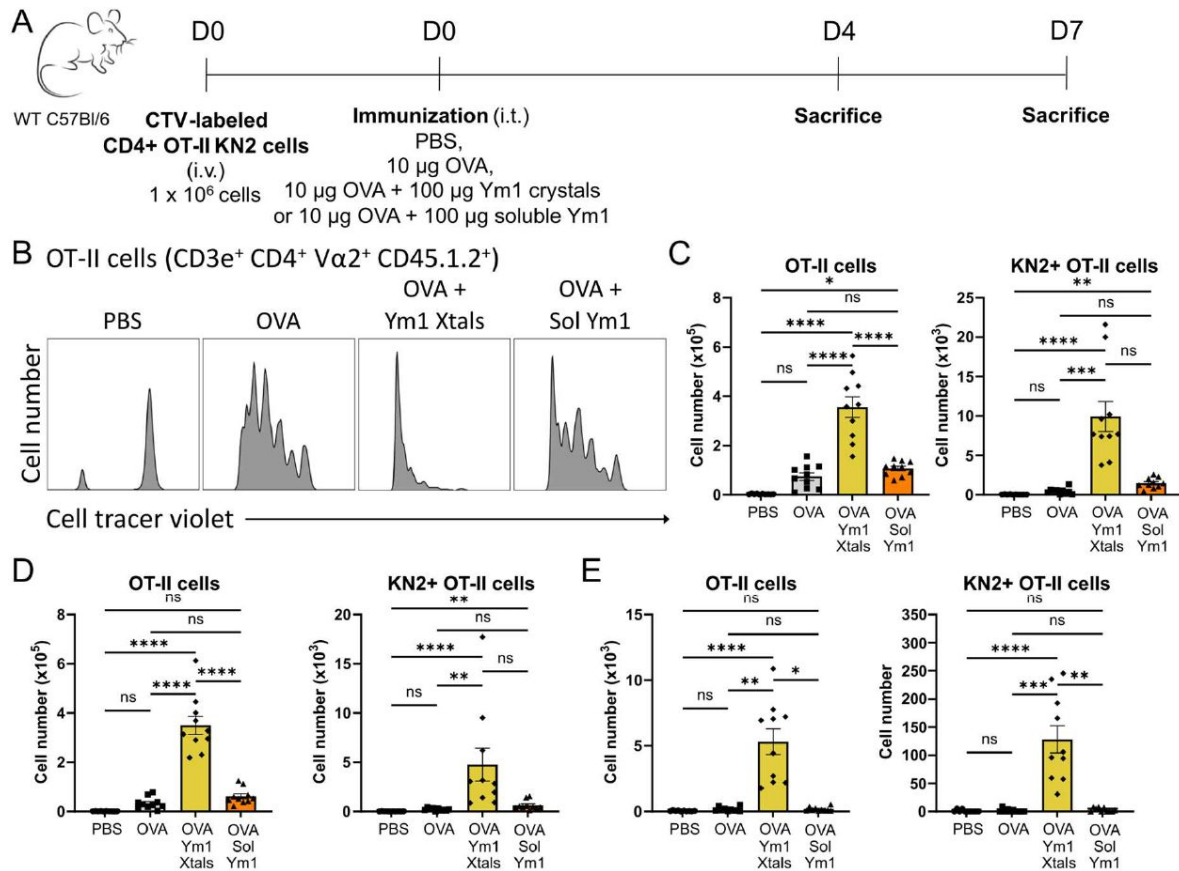


Figure 4

Ym1 crystals boost adaptive cellular immunity.

A) Scheme representing the used adoptive transfer model. **B-E**) Proliferation and KN2 expression of OT-II T cells in the mLNs and lungs 4 and 7 days after i.t. administration of PBS, OVA alone, OVA plus Ym1 crystals or OVA plus soluble Ym1. **(B)** Representative histograms of the division profile of the adoptively transferred OT-II T cells (CD3e⁺ CD4⁺ Vα2⁺ CD45.1.2⁺) in the mLNs on day 4. **(C, D)** The numbers of OT-II T cells and KN2+ OT-II T cells in the mLNs on day 4 **(C)** and day 7 **(D)**. **(E)** The numbers of OT-II T cells and KN2+ OT-II T cells in the lungs on day 7. Data are pooled from 2 independent experiments with n = 10 mice per group. Data are shown as mean ± SEM. ns P ≥ 0.05; *P < 0.05; **P < 0.01; ***P < 0.001; ****P < 0.0001.

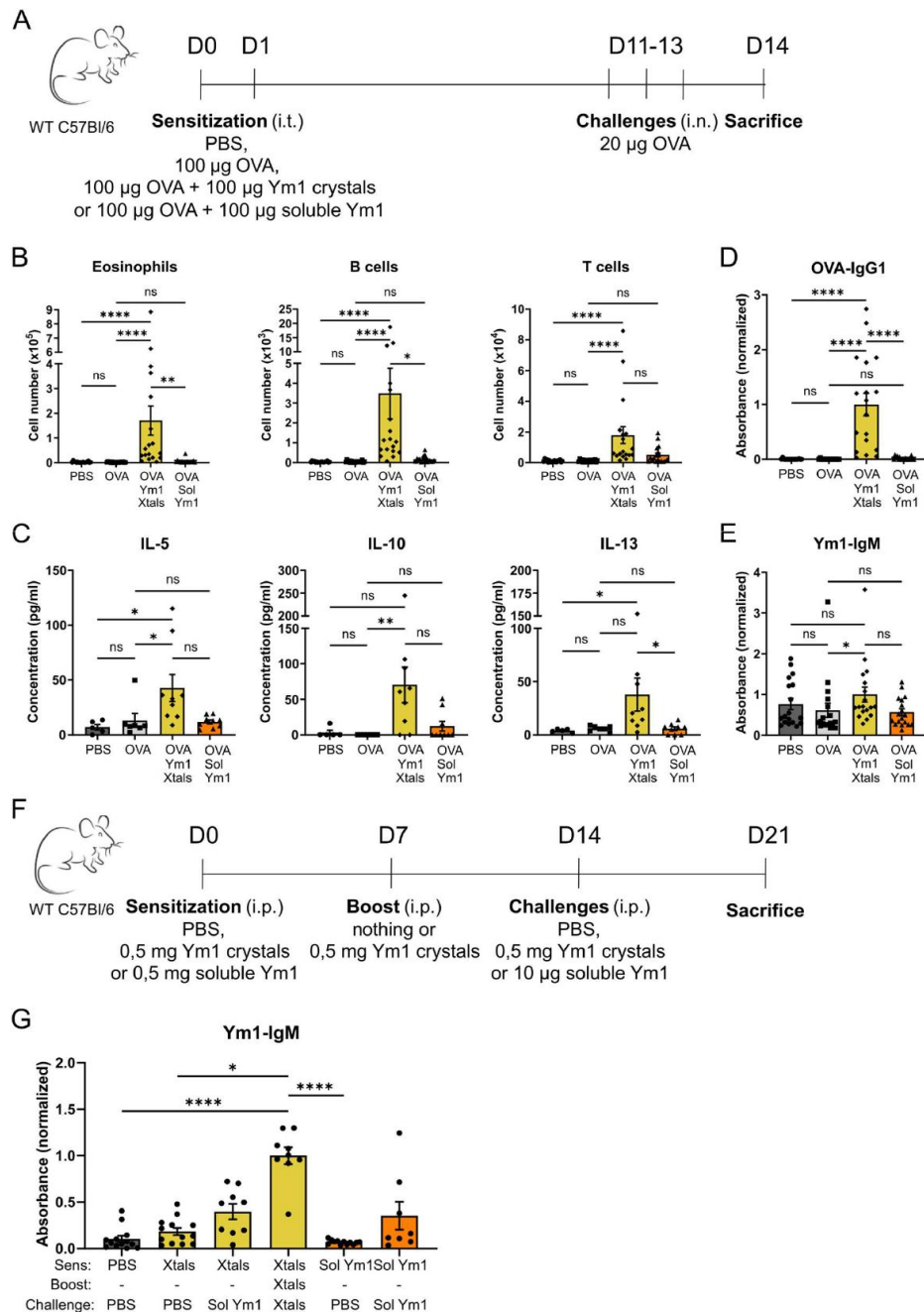


Figure 5

Ym1 crystals can act as type 2 adjuvant in OVA-induced asthma model and activate adaptive humoral immunity.

A) Scheme representing the used OVA-induced asthma model. **B)** Flow cytometric analysis of the immune cell influx in the BAL fluid of mice sensitized i.t. with PBS, OVA alone, OVA plus Ym1 crystals or OVA plus soluble Ym1 and challenged i.n. with OVA. Data are pooled from 2 independent experiments with $n = 18$ per group. **C)** Levels of IL-5, IL-10 and IL-13 in the supernatant of mLN cells after restimulation with OVA for 20 hours. Data are from 1 experiment with $n = 5, 7, 9$ and 9 , respectively. **D)** Serum OVA-specific IgG1 antibodies were measured by ELISA. **E)** Serum Ym1-specific IgM antibodies were measured by ELISA. Data are pooled from 2 independent experiments with $n = 18$ per group. **F)** Scheme representing the adaptive humoral immune response experiment. **G)** Serum Ym1-specific IgM antibodies were measured by ELISA. Data are pooled from 3 independent experiments with $n = 13, 13, 9, 9, 12$ and 8 , respectively. Data are shown as mean $\pm$ SEM. ns $P \geq 0.05$; * $P < 0.05$; ** $P < 0.01$; **** $P < 0.0001$. If pairwise comparison is not shown, the result is not significant.

As we were unable to make a crystallization-deficient soluble Ym1 mutein, we used WT soluble Ym1 protein as a control. However, we took into consideration that soluble Ym1 might crystallize *in vivo* after i.t. injection. To test this, we stained BAL pellets using a polyclonal anti Ym1/2 antibody and analyzed the samples using the ImageStream flow cytometry platform. The results clearly showed that the i.t. injected pre-formed Ym1 crystals were still present as intact crystals after 6 and 24 hours. The soluble Ym1 protein indeed crystallized *in vivo*, with crystals being clearly detectable after 6 hours and accumulating further after 24 hours. The crystals formed *in vivo* after i.t. injection of soluble Ym1 were also plate-like, but had a less uniform distribution and were smaller in size than the i.t. injected Ym1 crystals (**Fig. 2 F**). The total number of crystals found in the BAL pellet of soluble Ym1 treated mice (266 crystals in 5 pooled BAL pellets) was lower than in the BAL pellet of Ym1 crystal treated mice (1355 crystals in 5 pooled BAL pellets), potentially explaining why the soluble Ym1 group had less inflammation compared with the group receiving pre-formed Ym1 crystals.

Ym1 crystals activate DCs in the lymph nodes

We next studied whether innate immunity triggered by pseudo-CLCs was accompanied by the induction of adaptive immunity, particularly since IL-1 β , IL-33, IL-6 and TNF α in the lungs can activate antigen presenting cells⁴². Airway DCs bridge innate and adaptive immunity and can be activated by recombinant Gal10 crystals and UA crystals^{34,41}. Different subsets of DCs can be clearly distinguished from recruited monocytes and resident macrophages based on the expression of CD26 and lack of expression of the complement C5a receptor CD88, and further separated into XCR1⁺ cDC1 and CD172 (SIRP α)⁺ cDC2 (see **Fig. 3A** for gating strategy)^{30,43}. In mice given Ym1 crystals in the presence of fluorescently-labeled OVA antigen (**Fig. 3 B**), there was a clear increase in the number of total lung-draining mLN CD26⁺CD88⁺CD11c⁺CD64⁺ conventional DCs (cDCs), numerically mainly caused by an increase in the number of CD172⁺ cDC2, compared with mice receiving PBS or soluble Ym1, although XCR1⁺ cDC1s also clearly reacted to the crystals. There was also an increase in the number of mLN CD88⁺CD11b⁺ monocytes that expressed intermediate levels of CD11c induced by crystals (**Fig. 3 C**). Not surprisingly, the exposure to crystalline Ym1 also led to an increase in cDC1s, cDC2s and monocytes taking up the co-administered fluorescent OVA antigen (**Fig. 3 D**). Exposure to soluble Ym1 and Ym1 crystals also induced an upregulation of the costimulatory molecules CD80 (data not shown) and CD86, and this effect was most pronounced with crystalline Ym1, and particularly in cDCs that were also taking up fluorescent OVA (**Fig. 3 E**).

Ym1 crystals boost adaptive type 2 immunity

We next asked whether crystalline or soluble Ym1 could boost adaptive cellular immunity to co-administered OVA, when DCs that have taken up antigen and matured migrate to mLNs (**Fig. 4 A**). To evaluate induction of type 2 immunity in naïve antigen-specific T cells directly, cell tracer violet labeled OVA-reactive CD4⁺ T cell receptor (TCR) MHCII-restricted transgenic T cells (OT-II T cells) crossed to the huCD2-IL-4 (KN2) reporter strain were transferred into WT mice, allowing us to track T cell division simultaneously with the acquisition of a Th2-polarized state *in vivo*⁴⁴. When innocuous OVA was injected i.t. together with Ym1 crystals, the adoptively transferred OT-II KN2 T cells showed increased proliferation 4 days (**Fig. 4 B**) and 7 days (data not shown) later in the mLNs, an effect absent when OVA was given with soluble Ym1. This led to the accumulation of higher numbers of OT-II T cells and Th2 polarizing KN2⁺ OT-II T cells in the mLNs 4 days (**Fig. 4 C**) and 7 days (**Fig. 4 D**) after the i.t. administration. After 7 days, divided OT-II T cells and KN2⁺ OT-II T cells could also be found in the lungs of mice receiving OVA mixed with Ym1 crystals (**Fig. 4 E**). These cells lost CD62L expression while expressing CD44, showing they had become effector cells (data not shown). These effects on T cell activation and Th2 polarization were again not observed when OVA was injected together with soluble Ym1. Early IL-4 production in OVA-reactive OT-II cells could also point to differentiation of T cells into T_{FH} cells that provide help for germinal center reactions. However, we did not add CXCR5 or PD1 to study T_{FH} development in greater detail.

We finally tested whether Ym1 crystals could also promote allergic type 2 inflammation to inhaled harmless antigens, typically observed in asthmatics. WT C57BL/6 mice were sensitized i.t. with OVA mixed with Ym1 crystals on days 0 and 1 and received OVA i.n. challenges on days 11 to 13 (**Fig. 5 A**). These mice exhibited airway inflammation characterized by an influx of eosinophils, B cells and T cells (**Fig. 5 B**), a higher production of the type 2 cytokines IL-5, IL-10 and IL-13 in the supernatants of mLN restimulation cultures (**Fig. 5 C**) and a robust OVA-specific IgG1 response, a hallmark of type 2 immunity, most likely driven by enhanced IL-4 production in Th2 or T_{FH} cells (**Fig. 5 D**). These effects were not observed when OVA alone was used as the immunogen or when OVA was co-administered with soluble Ym1. Thus, the induction of allergic type 2 inflammation was specifically related to the crystalline state of Ym1. In contrast to recombinant Gal10 crystals, which mounted high titers of Gal10-specific IgG1 antibodies when used as adjuvant in the same experimental set-up³⁴, Ym1 crystals did not elicit an antibody response directed to Ym1 (**Fig. 5 E**). However, when mice received repetitive injections of Ym1 crystals in a different set-up via i.p. injection (**Fig. 5 F**), high titers of Ym1-specific IgM were detected (**Fig. 5 G**), suggesting that Ym1 crystals can trigger an auto-immune response or elicit a recall response from natural IgM antibody producing peritoneal B cells.

Discussion

Ym1 and Ym2 are among the most strongly induced proteins in the airways of mice after allergen exposure^{45–49} and both proteins readily form pseudo-CLCs *in vivo*. As a result, Ym1 and Ym2 have become markers of type 2 inflammation. Here we show that the phase transition of soluble Ym1/Ym2 to crystalline pseudo-CLC states leads to profound immunostimulatory effects on innate and adaptive type 2 immunity. Sutherland and colleagues have shown that vector-based overexpression of Ym1 or Ym2 in the lung increased the number of neutrophils in the BAL fluid. Moreover, blocking antibodies directed against Ym1/2 also suppressed neutrophilic inflammation in the lungs of asthmatic mice and helminth-infected mice⁵⁰. Whether these observed effects were due to the crystalline or soluble state of Ym1/2 remains unclear from these studies. Also in the present study, we lacked a proper soluble Ym1 control, since ImageStream analysis showed the soluble Ym1 protein to readily crystallize upon injection in the lungs. So far, several attempts to produce crystallization-deficient soluble Ym1 or Ym2 muteins failed, despite the fact that this was feasible for the CLC protein Gal10³⁴. Antibodies to Gal10 also dissolved pre-existing CLCs, but from the published literature it is unclear if the Ym1/Ym2 antibodies used would also suppress lung neutrophilia by dissolving Ym1/2 pseudo-CLC^{34,50}. We also observed that Ym1 crystals induced an auto-antibody response of the IgM isotype, suggesting intrinsic adjuvant activity in the crystalline state of Ym1. Antibodies to Ym1/2 have been described to occur also in a model of incipient neoplasia in mice overexpressing the Epstein Barr virus oncogene LMP1 in the skin, leading to chronic inflammation⁵¹. However, it is unclear if induction of such auto-antibodies to Ym1/2 would be the result of spontaneous crystallization of Ym1/2 in the skin of these mice. Recognition of endogenously produced UA or cholesterol crystals by IgM has been described previously^{52,53}, and in the case of UA, it was even proposed that IgM is required to turn soluble UA into an immunostimulatory adjuvant⁵³. Future studies will have to establish if IgM contributes in some way to the adjuvant activity of Ym1 crystals or facilitates the transition of soluble to crystalline Ym1.

Ym1 (*Chil3*) and Ym2 (*Chil4*) share 95% nucleotide sequence identity and 91% amino acid sequence identity and we now show here that native *in vivo* grown Ym1 crystals are structurally similar to native *in vivo* grown Ym2 crystals. Thus, Ym2 crystals most likely will stimulate innate and adaptive immunity and act as a type 2 adjuvant, similar to Ym1 crystals, although we did not formally demonstrate this experimentally.

Understanding the requirements for protein crystal formation during type 2 immunity, studying the mechanisms whereby protein crystals stimulate type 2 immunity and having the proper tools to be able to distinguish between the soluble and the crystalline state will help us to better understand if targeting CLCs in human asthma and CRSwNP would be of value. Since CLCs and pseudo-CLCs are almost exclusively found in eosinophilic airway inflammation across many species, and share many biological effects, it appears that protein crystallization is a more generalized feature of type 2 immunity. The convergent evolution of CLCs and pseudo-CLCs as protein crystals in type 2 immunity strongly suggests a selective advantage of these protein crystals in the enhancement of type 2-driven immune responses.

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Protein complex	Recombinant Ym1	Ex vivo Ym1	Recombinant Ym2	Ex vivo Ym2
Data collection				
X-ray source	SOLEIL	PETRA III	PETRA III	SLS
(beamline)	(Proxima 1)	(P14)	(P14)	(PXL)
Wavelength (Å)	0.97857	0.9763	0.9763	1.0000089224
Space group	<i>P21</i>	<i>P21</i>	<i>P21</i>	<i>P21</i>
Cell dimensions				
<i>a</i> , <i>b</i> , <i>c</i> (Å)	50.33, 60.01, 60.12	50.52, 60.25, 60.12	51.09, 59.90, 59.88	51.16, 60.21, 60.17
α , β , γ (°)	90.00, 94.66, 90.00	90.00, 94.48, 90.00	90.00, 94.14, 90.00	90.00, 94.24, 90.00
Resolution (Å)	50.00-1.79 (1.90-1.79)	59.93-1.42 (1.50-1.42)	59.90-1.17 (1.24-1.17)	42.50-1.70 (1.81-1.70)
<i>R</i> _{meas} (%)	24.7 (112.1)	12.2 (138.2)	8.0 (35.8)	17.7 (136.4)
$\langle I/\sigma \rangle$	5.28 (1.54)	8.99 (1.19)	15.12 (4.74)	6.59 (0.96)
CC ½ (%)	97.9 (56.6)	99.7 (39.5)	99.9 (92.5)	99.2 (35.8)
Completeness (%)	99.2 (96.0)	98.1 (97.3)	98.4 (95.9)	97.7 (97.3)
Redundancy	4.1 (3.9)	3.3 (3.3)	6.8 (6.1)	3.5 (3.4)
ISa score	8.04	26.18	27.31	24.20
Wilson B (Å²)	20.318	20.936	12.048	24.584
Refinement				
Resolution (Å)	42.40-1.79	26.91-1.42	37.45-1.17	16.78-1.71
No. reflections	33 407	66 944	119 181	39 026
<i>R</i> _{work} / <i>R</i> _{free} (%)	17.29 / 19.86	18.10 / 20.52	15.75 / 16.83	17.71 / 20.48
No. non-H atoms				
Protein	2 957	2 957	2 976	2 970
Ligand/ion	10	4	24	6
Water	215	163	584	239
<i>B</i> -factors (Å²)				
Protein	12.09	15.58	10.28	18.50
Ligand/ion	21.52	30.73	16.20	30.69
Water	21.74	22.78	27.02	27.93
R.m.s. deviations				
Bond lengths (Å)	0.008	0.010	0.010	0.010
Bond angles (°)	0.91	0.98	1.07	0.97

§Each dataset was collected from a single crystal. *Values in parentheses are for highest-resolution shell.

Table S1.

Crystallographic data and refinement statistics^{§,*}

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Editors

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Reviewer #1 (Public Review):

Summary: The manuscript by Heyndrickx et al describes protein crystal formation and function that bears similarity to Charcot-Leyden crystals made of galectin 10, found in humans under similar conditions. Therefore, the authors set out to investigate CLP crystal formation and their immunological effects in the lung. The authors reveal the crystal structure of both Ym1 and Ym2 and show that Ym1 crystals trigger innate immunity, activated dendritic cells in the lymph node, enhancing antigen uptake and migration to the lung, ultimately leading to induction of type 2 immunity.

Strengths:

We know a lot about expression levels of CLPs in various settings in the mouse but still know very little about the functions of these proteins, especially in light of their ability to form crystal structures. As such data presented in this paper is a major advance to the field.

Resolving the crystal structure of Ym2 and the comparison between native and recombinant CLP crystals is a strength of this manuscript that will be a very powerful tool for further evaluation and understanding of receptor, binding partner studies including the ability to aid mutant protein generation.

The ability to recombinantly generate CLP crystals and study their function in vivo and ex vivo has provided a robust dataset whereby CLPs can activate innate immune responses, aid activation and trafficking of antigen presenting cells from the lymph node to the lung and further enhances type 2 immunity. By demonstrating these effects the authors directly address the aims for the study. A key point of this study is the generation of a model in which crystal formation/function an important feature of human eosinophilic diseases, can be studied utilising mouse models. Excitingly, using crystal structures combined with understanding the biochemistry of these proteins will provide a potential avenue whereby inhibitors could be used to dissolve or prevent crystal formation in vivo.

The data presented flows logically and formulates a well constructed overall picture of exactly what CLP crystals could be doing in an inflammatory setting in vivo. This leaves open a clear and exciting future avenue (currently beyond the scope of this work) for determining whether targeting crystal formation in vivo could limit pathology.

Weaknesses:

Although resolving the crystal structure of Ym2 in particular is a strength of the authors work, the weaknesses are that further work or even discussion of Ym2 versus Ym1 has not been directly demonstrated. The authors suggest Ym2 crystals will likely function the same as Ym1, but there is insufficient discussion (or data) beyond sequence similarity as to why this is the case. If Ym1 and Ym2 crystals function the same way, from an evolutionary point, why do mice express two very similar proteins that are expressed under similar conditions that can both crystallise and as the authors suggest act in a similar way. Some discussion around these points would add further value.

Additionally, the crystal structure for Ym1 has been previously resolved (Tsai et al 2004, PMID 15522777) and it is unclear whether the data from the authors represents an advance in the 3D structure from what is previously known.

Whilst also generating a model to understand charcot-leyden crystals (CLCs), the authors fail to discuss whether crystal shape may be an important feature of crystal function. CLCs are typically needle like, and previous publications have shown using histology and TEM that Ym1 crystals are also needle like. However, the crystals presented in this paper show only formation of plate like structures. It is unclear whether these differences represent different methodologies (ie histology is 2D slides), or differences in CLP crystals that are intracellular

versus extracellular. These findings highlight a key question over whether crystal shape could be important for function and has not been addressed by the authors.

Ym1/Ym2 crystals are often observed in conditions where strong eosinophilic inflammation is present. However, soluble Ym1 delivery in naïve mice shows crystal formation in the absence of a strong immune response. There is no clear discussion as to the conditions in which crystal formation occurs in vivo and how results presented in the paper in terms of priming or exacerbating an immune responses align with what is known about situations where Ym1 and Ym2 crystals have been observed.

Reviewer #2 (Public Review):

Summary: This interesting study addresses the ability of Ym1 protein crystals to promote pulmonary type 2 inflammation in vivo, in mice.

Strengths: The data are extremely high quality, clearly presented, significantly extending previous work from this group on the type 2 immunogenicity of protein crystals.

Weaknesses: There are no major weaknesses in this study. It would be interesting to see if Ym2 crystals behave similarly to Ym1 crystals in vivo. Some additional text in the Introduction and Discussion would enrich those sections.